








# Time Series Analysis

## Seminar 1: Stochastic processes and stationarity




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# Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **BET**: Bucharest Exchange Trading (index)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **EODHD**: EOD Historical Data (market-data provider)
- **GDP**: Gross Domestic Product
- **HICP**: Harmonised Index of Consumer Prices
- **MA**: Moving Average
- **PACF**: Partial AutoCorrelation Function

## Today's Question and Seminar Structure

- The question of the seminar
  - **is a series stationary, is it white noise, and how do the ACF and the Ljung–Box test answer these questions?**
- Results to obtain (nothing is handed in)
  - the answers to the A exercises
  - a figure and a table for each B exercise
  - the corrected AI answer of C2
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

# Exercise map

| Task | Question | Type | Model |
|---|---|---|---|
| A1, A2 | mean, autocovariances and ACF of MA(1) and MA(2) processes | Solved, Proposed | A1 |
| A3, A4 | a random walk; a trend-stationary series and its first difference | Solved, Proposed | A3 |
| A5, A6 | sample ACF, bands, Box–Pierce and Ljung–Box by hand | Solved, Proposed | A5 |
| B1, B2 | prices and returns: BET; S&P 500 and EUR/RON | Solved, Proposed | B1 |
| B3, B4 | transformations: Romanian real GDP; Romanian HICP inflation | Solved, Proposed | B3 |
| C1, C2 | has inflation become more persistent? what is wrong in an AI answer? | Proposed | B1, B4 |

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in class
- Part A: calculations on paper; Part B: real data in the notebook, each task ending with an interpretation question; Part C: open questions and the critique of an AI answer

# Data used

| Series | Source | Frequency | Period |
|---|---|---|---|
| BET, S&P 500 | EODHD | daily, close | 2000–2026 |
| EUR/RON | BNR reference rate | daily | 2005–2026 |
| Real GDP, Romania | Eurostat (namq_10_gdp) | quarterly, not adjusted | 1995–2026 |
| HICP, Romania | Eurostat (prc_hicp_minr) | monthly, 2015 = 100 | 1996–2026 |

- Daily log returns in %: $r_t = 100\,(\ln P_t - \ln P_{t-1})$; weekends and repeated holiday closes are dropped; the last day is 18 September 2026
- Growth and inflation rates in %: log differences over one period or one year (Prerequisites 13/15)
- In the notebook: `b1_returns('bet')`, `sample_acf(x)`, `ljung_box(x, 10)`; no account or key is needed

# Prerequisites for Today (1/15): processes and moments

- **Stochastic process** $\{X_t\}$: one random variable for each date $t$
  - a time series $x_1, \ldots, x_T$ is one observed path of the process
  - $E[\cdot]$: the expected value, an average over all possible paths at a fixed date
- **Mean** and **autocovariance**

$$\mu_t = E[X_t], \qquad \gamma(t,s) = \mathrm{Cov}(X_t, X_s) = E[(X_t - \mu_t)(X_s - \mu_s)]$$

  - $\gamma(t,s) > 0$: when $X_t$ is above its mean, $X_s$ tends to be above its mean too; $\gamma(t,t) = \mathrm{Var}(X_t)$
  - correlation: $\mathrm{Corr}(X_t, X_s) = \gamma(t,s)/\sqrt{\mathrm{Var}(X_t)\,\mathrm{Var}(X_s)}$, between $-1$ and $1$
- **Rules for covariances**, with constants $a, b, c$
  - $\mathrm{Cov}(aX + bY, Z) = a\,\mathrm{Cov}(X,Z) + b\,\mathrm{Cov}(Y,Z)$; $\mathrm{Cov}(X + c, Y) = \mathrm{Cov}(X,Y)$
  - $\mathrm{Var}(\sum_i a_i X_i) = \sum_i \sum_j a_i a_j \mathrm{Cov}(X_i, X_j)$; for uncorrelated $X_i$: $\sum_i a_i^2 \mathrm{Var}(X_i)$

# Prerequisites for Today (2/15): stationarity

- **Weakly stationary**: finite variance, constant mean, autocovariance that depends only on the lag

$$E[X_t^2] < \infty, \qquad E[X_t] = \mu, \qquad \text{Cov}(X_t, X_{t+h}) = \gamma(h) \quad \text{for every } t$$

  - $h$: the lag, the distance in periods between two dates; $\gamma(0) = \text{Var}(X_t)$
  - a shift in time does not change the mean, the variance or the covariances
- **ACF** (autocorrelation function): the autocovariance in units of the variance

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}, \qquad \rho(0) = 1, \quad |\rho(h)| \leq 1, \quad \rho(-h) = \rho(h)$$

- **Strictly stationary**: every joint distribution is unchanged by a shift in time
  - $(X_{t_1}, \ldots, X_{t_k})$ and $(X_{t_1+h}, \ldots, X_{t_k+h})$ have the same distribution, for every $k$, dates $t_1, \ldots, t_k$ and shift $h$
  - **i.i.d.** (independent and identically distributed): the $X_t$ are independent and all have the same distribution

# Prerequisites for Today (3/15): white noise and moving averages

- **White noise** $\varepsilon_t \sim \mathrm{WN}(0, \sigma^2)$: unpredictable shocks

$$E[\varepsilon_t] = 0, \qquad \mathrm{Var}(\varepsilon_t) = \sigma^2, \qquad \mathrm{Cov}(\varepsilon_t, \varepsilon_s) = 0 \ \ (t \neq s)$$

  - $\sigma^2$: the variance of a shock
  - weak (only uncorrelated), i.i.d. (independent), Gaussian (i.i.d. Normal)
- **MA($q$)** (moving average): a weighted sum of the current and the last $q$ shocks

$$X_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

  - $\mu$: the mean; $\theta_1, \ldots, \theta_q$: the weights of the past shocks; $q$: the order
- **Autocovariances of an MA($q$)**: only the shocks common to $X_t$ and $X_{t+h}$ count

$$\gamma(h) = \sigma^2 \sum_{j=0}^{q-h} \theta_j \theta_{j+h} \ \ (0 \leq h \leq q), \qquad \gamma(h) = 0 \ \ (h > q)$$

  - with $\theta_0 = 1$; $X_t$ and $X_{t+h}$ share the shocks $\varepsilon_{t+h-q}, \ldots, \varepsilon_t$
  - beyond lag $q$ no shock is shared: the ACF cuts off after lag $q$

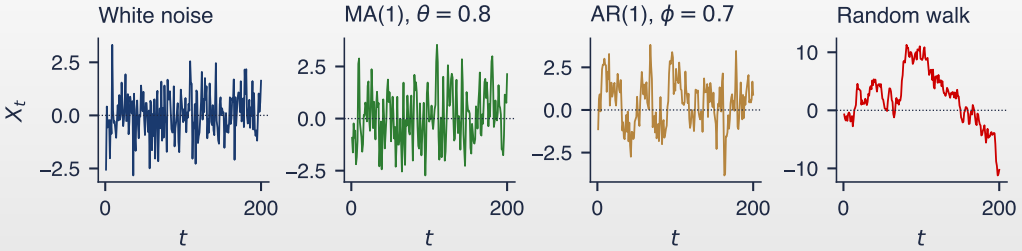
# Prerequisites for Today (4/15): random walk, AR(1) and trends

- **Random walk**: each value is the previous one plus a shock

$$X_t = X_{t-\mathbf{1}} + \varepsilon_t, \quad X_{\mathbf{0}} = 0 \implies X_t = \varepsilon_{\mathbf{1}} + \cdots + \varepsilon_t$$

  - with drift: $X_t = c + X_{t-\mathbf{1}} + \varepsilon_t$; $c$: the drift, the average change per period
  - every shock stays in all later values
- **AR(1)** (autoregressive): $X_t = \phi X_{t-\mathbf{1}} + \varepsilon_t$, with $|\phi| < 1$
  - $\phi$: the share of today's value that passes to tomorrow; $\rho(h) = \phi^h$, a geometric decay
  - $\phi = 1$ gives the random walk
- **Trend-stationary**: $Y_t = a + bt + u_t$, with $u_t$ stationary
  - $a + bt$: a deterministic line; $b$: its slope, the average change per period
- **Differences**
  - $\Delta X_t = X_t - X_{t-\mathbf{1}}$; seasonal: $\Delta_s X_t = X_t - X_{t-s}$, $s$: the number of periods in a year

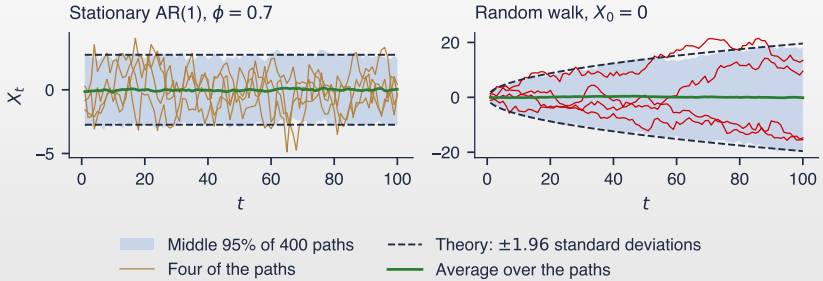
# Prerequisites for Today (5/15): one path of four processes




- Simulated with the same kind of shocks, $\varepsilon_t \sim N(0,1)$, $T = 200$
- From left to right, more and more memory
  - white noise: no pattern; MA(1): neighbours are similar, the memory lasts one period
  - AR(1) with $\phi = 0.7$: longer waves, but always back around 0
  - random walk: no level to return to; the path wanders far from 0

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# Prerequisites for Today (6/15): many paths at once




- 400 simulated paths of each process; blue band: the middle 95% of the paths at each date $t$
- The mean and the variance at a fixed $t$ are computed across the paths
  - AR(1): the band has the same width at every $t$, $\pm 1.96\sqrt{1/(1-\phi^2)} = \pm 2.74$: stationary
  - random walk: the band widens like $\pm 1.96\sqrt{t}$; the variance grows with $t$: not stationary

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# Prerequisites for Today (7/15): the sample ACF and its band

- **Sample autocovariance** and **sample ACF** (divisor $T$)

$$\hat{\gamma}(h) = \frac{1}{T}\sum_{t=1}^{T-h}(x_t - \bar{x})(x_{t+h} - \bar{x}), \qquad \hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}$$

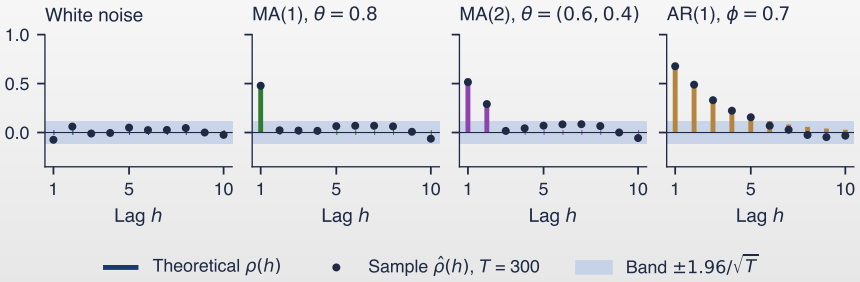
  - $\bar{x} = \frac{1}{T}\sum_t x_t$: the sample mean; the hat marks a value estimated from the data
  - the sample ACF estimates $\rho(h)$ only if the series is stationary

- **Under i.i.d. data**: $\hat{\rho}(h) \approx N(0, 1/T)$, with standard error $1/\sqrt{T}$; the **95% band** is

$$\pm\frac{1.96}{\sqrt{T}}$$

  - 1.96: the 97.5% quantile of the standard Normal distribution
  - a bar outside the band: $\rho(h) \neq 0$ at 5%, for that lag alone
  - with 20 lags, about 1 bar crosses the band by chance

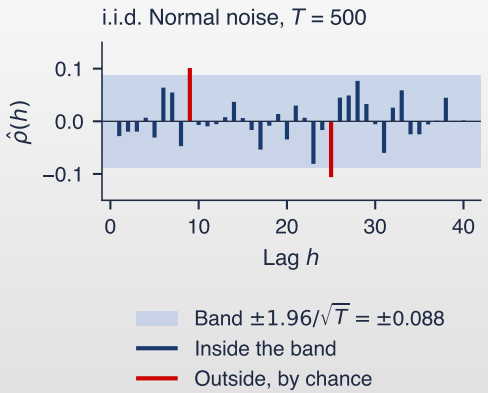
# Prerequisites for Today (8/15): theoretical and sample ACF




- Bars: the theoretical $\rho(h)$; dots: $\hat{\rho}(h)$ from one simulated path, $T = 300$
- Typical patterns
  - white noise: no bar; MA(1): one bar, $\rho(1) = \theta/(1+\theta^2) = 0.49$; MA(2): two bars, then nothing
  - AR(1): $\rho(h) = 0.7^h$, a geometric decay
  - random walk or trend: a very slow, almost linear decay; seasonal data: peaks at lags $s, 2s, \ldots$

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# Prerequisites for Today (9/15): the band on i.i.d. noise




- One simulated i.i.d. Normal series, $T = 500$: band $\pm 1.96/\sqrt{500} = \pm 0.088$
- 2 of the 40 bars (lags 9 and 25) cross the band
  - expected: 5% of 40 lags is 2 lags
  - a single bar just outside the band is weak evidence; the portmanteau tests look at all lags together
- **Significant is not large**
  - $\hat{\rho}(1)^2$: about the share of the variance of $x_t$ that a linear forecast from $x_{t-1}$ explains
  - with $T = 5\,000$, $\hat{\rho}(1) = 0.03$ is outside the band, yet explains only 0.09% of the variance

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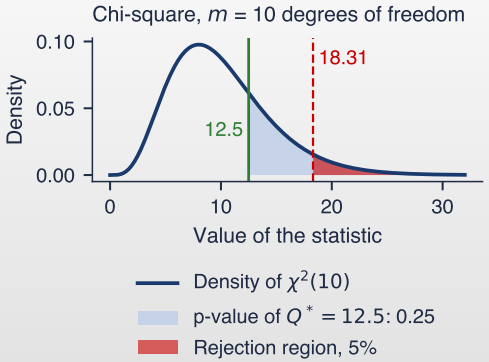
# Prerequisites for Today (10/15): portmanteau tests

- $H_0$: $\rho(1) = \cdots = \rho(m) = 0$, the series is white noise up to lag $m$
  - $m$: the number of lags tested; $H_1$: at least one $\rho(h) \neq 0$
- **Box–Pierce** and **Ljung–Box**: sums of squared sample autocorrelations

$$Q(m) = T \sum_{h=1}^{m} \hat{\rho}(h)^2, \qquad Q^*(m) = T(T+2) \sum_{h=1}^{m} \frac{\hat{\rho}(h)^2}{T-h}$$

  - squares: positive and negative autocorrelations both raise the statistic
  - the factor $(T+2)/(T-h) > 1$ corrects the small-sample behaviour of $Q$
  - under $H_0$ both are approximately $\chi^2(m)$, the chi-square distribution with $m$ degrees of freedom
- **Decision**: reject $H_0$ at 5% if the statistic exceeds $\chi^2_{0.95}(m)$, the 95% quantile of $\chi^2(m)$
  - 5% critical values: $\chi^2_{0.95}(2) = 5.99$, $\chi^2_{0.95}(3) = 7.81$, $\chi^2_{0.95}(8) = 15.51$, $\chi^2_{0.95}(10) = 18.31$, $\chi^2_{0.95}(12) = 21.03$

# Prerequisites for Today (11/15): critical value and p-value



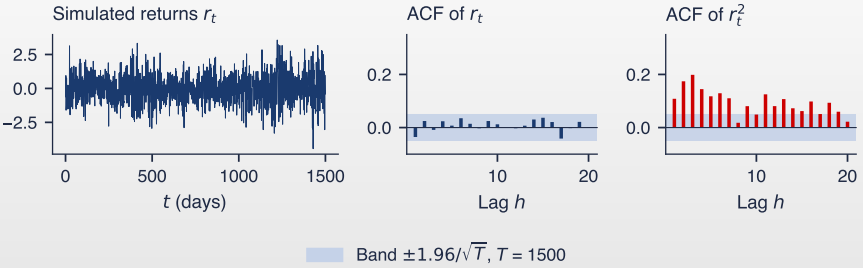

- **p-value**: the probability, under $H_0$, of a statistic at least as large as the observed one

$$p = P(\chi^2(m) \geq Q^*_{obs})$$

  - $Q^*_{obs}$: the value computed from the data; $p$: the area to the right of it
  - reject $H_0$ at 5% when $p < 0.05$, the same as $Q^*_{obs} > \chi^2_{0.95}(m)$
- **Chart**: $m = 10$
  - critical value 18.31: 5% of the area lies to its right (red)
  - $Q^*_{obs} = 12.5$ gives $p = 0.25$ (blue area): do not reject white noise
  - the mean of $\chi^2(m)$ is $m$: values near $m$ are what $H_0$ expects

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# Prerequisites for Today (12/15): volatility clustering: returns and their squares




- **Volatility clustering**: large moves, of either sign, come in groups
  - simulated daily returns $r_t$ whose variance depends on yesterday's squared return, $T = 1\,500$
- Two ACFs of the same series
  - $r_t$: $\hat{\rho}(1) = -0.04$, all bars near the band: the direction of a move is not predictable
  - $r_t^2$: $\hat{\rho}(1) = 0.11$, bars outside the band up to lag 20: the size of a move is predictable
  - so $r_t$ is uncorrelated, but not independent

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# Prerequisites for Today (13/15): transformations

- **Logarithm** of a positive series $Y_t$: $\ln Y_t$
  - stabilises a variance that grows with the level; a constant growth rate becomes a straight line
- **Growth rates** in % from the log

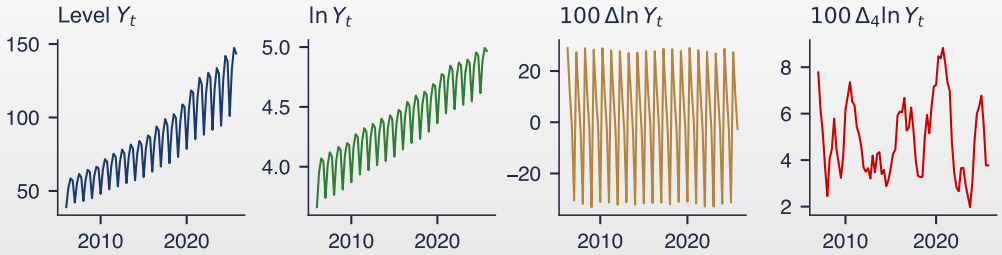
$$100\,\Delta \ln Y_t = 100\,(\ln Y_t - \ln Y_{t-\mathbf{1}}), \qquad 100\,\Delta_{\mathbf{4}} \ln Y_t = 100\,(\ln Y_t - \ln Y_{t-\mathbf{4}})$$

  - $\Delta$: growth over one period; $\Delta_{\mathbf{4}}$ ($\Delta_{\mathbf{12}}$): growth over a year for quarterly (monthly) data
  - differences remove trends and random walks; $\Delta_{\mathbf{4}}$ and $\Delta_{\mathbf{12}}$ also remove the season
- **Box–Cox**: a family of power transformations

$$Y_t^{(\lambda)} = \frac{Y_t^{\lambda} - 1}{\lambda} \quad (\lambda \neq 0), \qquad Y_t^{(\mathbf{0})} = \ln Y_t$$

  - $\lambda$: the power, chosen so that the variance becomes stable; $\lambda = 1$: no change in shape

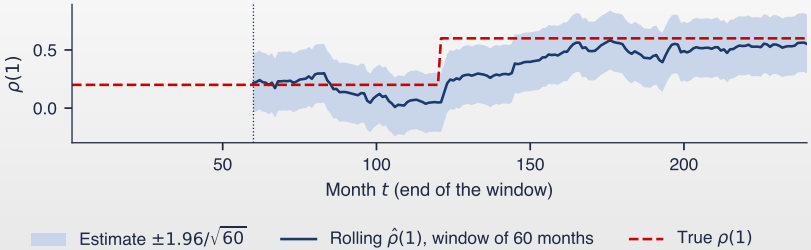
# Prerequisites for Today (14/15): transformations of a growing seasonal series




- A simulated quarterly series, 2006–2025, growing and seasonal; four views of the same data
- What each transformation does
  - level: the seasonal swing grows with the level; $\ln Y_t$: the swing has a constant size
  - $100\,\Delta \ln Y_t$: the trend is gone, the season dominates (standard deviation 21.9%)
  - $100\,\Delta_4 \ln Y_t$: trend and season are gone; mean 4.9%, standard deviation 1.7%

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# Prerequisites for Today (15/15): rolling-window estimates




- **Rolling window** of length $w$: the statistic is recomputed on the last $w$ observations at each date $t$
  - $\hat{\rho}_t(1)$: the sample $\hat{\rho}(1)$ of $x_{t-w+1}, \ldots, x_t$; it shows whether the persistence changes over time
- Chart: a simulated AR(1) whose $\phi$ jumps from 0.2 to 0.6 at month 120; $w = 60$
  - the estimate follows the change with a delay of up to $w$ months
  - standard error about $1/\sqrt{w} = 0.13$; shaded: the estimate $\pm 1.96$ standard errors

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# A1: an MA(1) process [Solved]

## Task

- $X_t = \varepsilon_t + 0.5\,\varepsilon_{t-\mathbf{1}}$, with $\varepsilon_t \sim \text{WN}(0, 2)$.
- 1. Compute the mean $E[X_t]$.
- 2. Compute $\gamma(0)$, $\gamma(1)$ and $\gamma(2)$.
- 3. Compute $\rho(1)$ and $\rho(2)$.
- 4. Say whether the process is weakly stationary, and why.
- Report: five numbers and one sentence.

## Solution

- 1. $E[X_t] = 0 + 0.5 \cdot 0 = 0$
- 2. $\gamma(0) = \text{Var}(\varepsilon_t) + 0.25\,\text{Var}(\varepsilon_{t-\mathbf{1}}) = 2 + 0.5 = 2.50$
- $\gamma(1) = \text{Cov}(\varepsilon_t + 0.5\varepsilon_{t-\mathbf{1}}, \varepsilon_{t+\mathbf{1}} + 0.5\varepsilon_t) = 0.5 \cdot 2 = 1.00$; $\gamma(2) = 0$ (no common shock)
- 3. $\rho(1) = 1.00/2.50 = 0.40$; $\rho(2) = 0$
- 4. Yes: the mean and all $\gamma(h)$ are finite and do not depend on $t$.

Back Exercise list

# A2: an MA(2) process [Proposed]

Solution discussed in class

## Task

- $X_t = 2 + \varepsilon_t + 0.4\,\varepsilon_{t-\mathbf{1}} - 0.3\,\varepsilon_{t-\mathbf{2}}$, with $\varepsilon_t \sim \text{WN}(0, 1)$. Model: A1.
- 1. Compute the mean.
- 2. Compute $\gamma(0)$, $\gamma(1)$, $\gamma(2)$ and $\gamma(3)$.
- 3. Compute $\rho(1)$ and $\rho(2)$, and sketch the ACF up to lag 4.
- 4. Say how the constant 2 changes the autocovariances.
- Report: six numbers, the sketch and one sentence.

Back Exercise list

# A3: a random walk [Solved]

## Task

- $X_t = X_{t-1} + \varepsilon_t$, $X_0 = 0$, $\varepsilon_t \sim \text{WN}(0, 0.25)$.
- 1. Compute $E[X_{100}]$, $\text{Var}(X_{100})$ and $\text{Var}(X_{120})$.
- 2. Compute $\text{Cov}(X_{100}, X_{120})$ and $\text{Corr}(X_{100}, X_{120})$.
- 3. Say whether $X_t$ and $\Delta X_t$ are stationary.
- 4. With a drift $c = 0.1$, compute $E[X_{100}]$.
- Report: six numbers and two sentences.

## Solution

- 1. $X_t = \sum_{i=1}^{t} \varepsilon_i$: $E[X_{100}] = 0$;
  $\text{Var}(X_{100}) = 100 \cdot 0.25 = 25$; $\text{Var}(X_{120}) = 30$
- 2. The common shocks are $\varepsilon_1, \ldots, \varepsilon_{100}$: $\text{Cov} = 25$;
  $\text{Corr} = 25/\sqrt{25 \cdot 30} = \sqrt{100/120} = 0.913$
- 3. $X_t$: no, its variance grows with $t$; $\Delta X_t = \varepsilon_t$: yes, it is white noise.
- 4. $E[X_{100}] = 100 \cdot 0.1 = 10$; the variance does not change.

Back Exercise list

# A4: a trend-stationary series [Proposed]

Solution discussed in class

## Task

- $Y_t = 5 + 0.2\,t + \varepsilon_t$, $\varepsilon_t \sim \mathrm{WN}(0, 1)$. Model: A3.
- 1. Compute $E[Y_t]$ and $\mathrm{Var}(Y_t)$, and say whether $Y_t$ is stationary.
- 2. Write $D_t = \Delta Y_t$ and compute its mean, $\gamma(0)$, $\gamma(1)$ and $\rho(1)$.
- 3. Compare $Y_t$ with a random walk with drift 0.2: what happens to a shock $\varepsilon_{\mathbf{50}}$ in each?
- Report: five numbers and two sentences.

# A5: a sample ACF by hand [Solved]

## Task

- Eight observations: $x = (4, 6, 5, 8, 7, 9, 6, 7)$.
- 1. Compute $\bar{x}$ and the deviations $x_t - \bar{x}$.
- 2. Compute $\hat{\gamma}(0)$, $\hat{\gamma}(1)$, $\hat{\gamma}(2)$ with divisor $T = 8$, then $\hat{\rho}(1)$ and $\hat{\rho}(2)$.
- 3. Compute the 95% band (Prerequisites 7/15) and compare.
- 4. Compute the Ljung–Box $Q^*(2)$ and decide at 5%.
- Report: seven numbers and one sentence.

## Solution

- 1. $\bar{x} = 52/8 = 6.5$; deviations $(-2.5, -0.5, -1.5, 1.5, 0.5, 2.5, -0.5, 0.5)$
- 2. sum of squares 18: $\hat{\gamma}(0) = 2.250$; products at lag 1 sum to 0.25: $\hat{\gamma}(1) = 0.0312$; at lag 2 to 7.00: $\hat{\gamma}(2) = 0.875$
- $\hat{\rho}(1) = 0.014$, $\hat{\rho}(2) = 0.389$
- 3. Band $\pm 0.693$: both inside
- 4. $Q^*(2) = 8 \cdot 10\,(\hat{\rho}(1)^2/7 + \hat{\rho}(2)^2/6) = 2.02 < 5.99$ ($p = 0.36$): do not reject white noise; with $T = 8$ the test has almost no power.

# A6: Box–Pierce or Ljung–Box? [Proposed]

Solution discussed in class

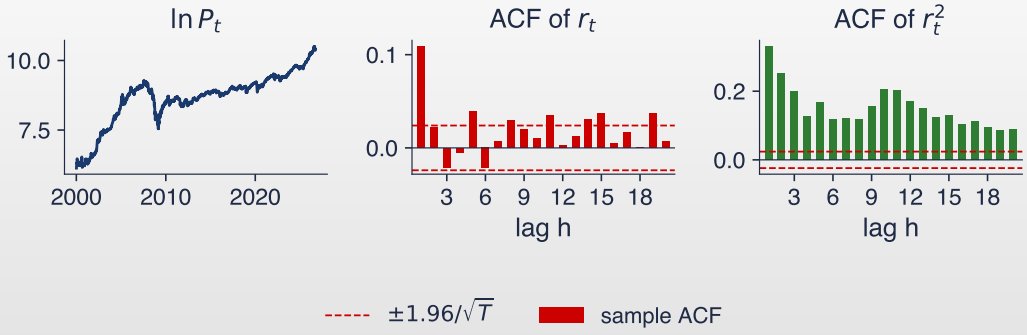
## Task

- A series of $T = 120$ monthly returns has $\hat{\rho}(1) = 0.21$, $\hat{\rho}(2) = -0.08$, $\hat{\rho}(3) = 0.12$. Model: A5.
- 1. Compute the band and say which autocorrelations lie outside it.
- 2. Compute $Q(3)$ and $Q^*(3)$.
- 3. Compare both with $\chi^2_{0.95}(3) = 7.81$ and decide.
- 4. Explain why the two tests can disagree, and which one you trust here.
- Report: four numbers and two sentences.

# B1: BET prices and returns [Solved]

- **Question**: is the BET log price stationary, and are BET daily returns white noise?
- **Inputs**: BET closes since 2000; $\ln P_t$ and $r_t = 100\,\Delta \ln P_t$
- **Tasks**
  1. Compute the sample ACF of $\ln P_t$ at lags 1, 10 and 50.
  2. Draw the ACF of $r_t$ up to lag 20 with the 95% band (Prerequisites 7/15), and count the bars outside it.
  3. Compute Ljung–Box $Q^*(10)$ for $r_t$ and for $r_t^2$, with their p-values.
  4. Interpretation: can yesterday's return be used to forecast today's return?
- **Report**: three ACF values, the chart, two statistics with p-values and two sentences
- Notebook: section B1

## B1: solution (1/2) [Solved]




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## B1: solution (2/2) [Solved]

- $\ln P_t$: $\hat{\rho}(1) = 0.999$, $\hat{\rho}(10) = 0.991$, $\hat{\rho}(50) = 0.953$: almost no decay, a random-walk-like level, not stationary
- $r_t$ ($T = 6\,670$, band $\pm 0.024$): 7 of 20 bars outside, the largest $\hat{\rho}(1) = 0.109$; $Q^*(10) = 108.0$, $p < 0.001$
- $r_t^2$: $\hat{\rho}(1) = 0.33$; $Q^*(10) = 2440$, $p < 0.001$: much stronger dependence
- Interpretation: yesterday's return explains only $\hat{\rho}(1)^2 = 1.2\%$ of today's variance, too little for a trading rule after costs; the size of yesterday's move does predict today's risk

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# B2: the S&P 500 and EUR/RON [Proposed]

- **Question**: do the S&P 500 and the EUR/RON rate behave like the BET?
- **Inputs**: S&P 500 closes since 2000; EUR/RON (BNR reference rate) since July 2005; model: B1
- **Tasks**
  1. Repeat the three steps of B1 for both series.
  2. Compare the sign and the size of $\hat{\rho}(1)$ of the returns across BET, S&P 500 and EUR/RON.
  3. Compare $Q^*(10)$ of the squared returns across the three series.
  4. Interpretation: which of the three return series is closest to white noise?
- **Report**: one table (three series, six numbers each) and two sentences
- Notebook: section B2

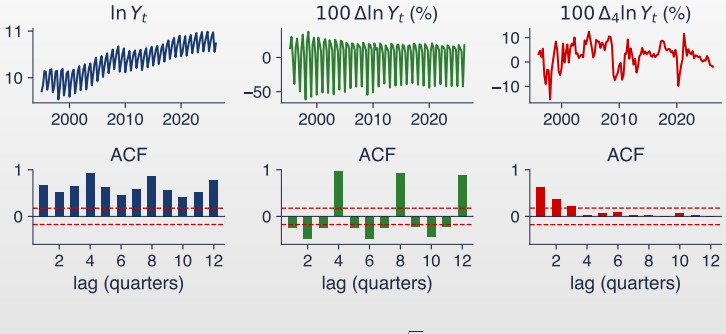
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# B3: Romanian real GDP [Solved]

- **Question**: which transformation makes Romanian quarterly GDP closest to stationary?
- **Inputs**: real GDP, chain-linked volumes (2010), not seasonally adjusted, Eurostat, from 1995
- **Tasks**
  1. Compute $\ln Y_t$, $100\,\Delta \ln Y_t$ and $100\,\Delta_4 \ln Y_t$.
  2. Draw the three series and their ACF up to lag 12.
  3. Report $\hat{\rho}(1)$, $\hat{\rho}(4)$ and $Q^*(8)$ for each.
  4. Interpretation: why is $\hat{\rho}(4)$ of the quarterly growth rate close to 1?
- **Report**: a table of nine numbers, the chart and two sentences
- Notebook: section B3

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# B3: solution (1/2) [Solved]




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# B3: solution (2/2) [Solved]

| | $\hat{\rho}(1)$ | $\hat{\rho}(4)$ | $Q^*(8)$ |
|---|---|---|---|
| $\ln Y_t$ | 0.68 | 0.94 | 495.0 |
| $100\,\Delta \ln Y_t$ | −0.24 | 0.96 | 329.6 |
| $100\,\Delta_4 \ln Y_t$ | 0.63 | 0.03 | 74.2 |

- $\ln Y_t$: trend and season, peaks at lags 4 and 8; $\Delta \ln Y_t$: the season dominates (standard deviation 27.5%); $\Delta_4 \ln Y_t$: mean 2.73%, the ACF dies out after 2–3 quarters
- Interpretation: a q/q rate of unadjusted data repeats the same seasonal swing every year, hence $\hat{\rho}(4) \approx 1$; the annual rate removes the season but reacts late to turning points and overlaps across quarters (positive $\hat{\rho}(1)$)

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## B4: Romanian inflation [Proposed]

- **Question**: is monthly Romanian inflation since 2005 white noise around its mean?
- **Inputs**: HICP, 2015 = 100, Eurostat, January 2005 to the last month available; model: B3
- **Tasks**
  1. Compute the monthly inflation $100\,\Delta \ln P_t$ and the 12-month inflation $100\,\Delta_{\mathbf{12}} \ln P_t$.
  2. Draw the ACF of both up to lag 36 and report $\hat{\rho}(1)$, $\hat{\rho}(12)$ and $\hat{\rho}(24)$.
  3. Compute Ljung–Box $Q^*(12)$ for the monthly inflation.
  4. Interpretation: what does the ACF of the monthly inflation say about how fast an inflation shock fades?
- **Report**: six ACF values, one statistic with its p-value, the chart and two sentences
- Notebook: section B4

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# C1: has Romanian inflation become more persistent? [Proposed]

- **Question**: is monthly inflation more persistent after 2020 than in 2005–2019?
- **Inputs**: monthly HICP inflation of B4, split in January 2020; models: B1, B4
- **Tasks**
  1. Compute $\hat{\rho}(1)$ and $Q^*(12)$ in each subperiod, with the band of each.
  2. Draw $\hat{\rho}(1)$ on rolling windows of 60 months.
  3. List the events of 2020–2024 that could change the persistence.
  4. Interpretation: why is the difference of two sample autocorrelations hard to judge with 80 observations?
- **Report**: a table, the chart and a plan for a project
- Notebook: section C1

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## C2: audit an AI answer [Proposed]

- A student asked an AI assistant to interpret the S&P 500 daily data since 2000 ($T = 6\,714$ returns). The answer:
- `(a) The ACF of the log price at lag 1 is 0.999, so the log price is a stationary process with a very long memory.`
- `(b) The return autocorrelation at lag 1 is −0.099, outside the band of 0.024, so a trader can forecast most of tomorrow's return.`
- `(c) Ljung-Box Q(10) of the squared returns is 6133, so the returns themselves are autocorrelated.`
- `(d) A weakly stationary process is always strictly stationary, since its mean and variance are constant.`
- `(e) Differencing a series that is already stationary does no harm: white noise stays white noise.`
- `(f) A random walk has Var(X(t)) = t * sigma^2, so it is not stationary.`
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - 2. Report: a list of six verdicts with one line of justification each.

## Key takeaways

- Autocovariances of linear processes follow from the covariance rules: only common shocks count
- A random walk has a growing variance; its difference is white noise; a trend-stationary series becomes an over-differenced MA(1)
- Look at the level, the changes and the squares: prices wander, returns are almost uncorrelated, squares are strongly correlated
- For macro data, choose the transformation first (log, $\Delta$, $\Delta_4$, $\Delta_{12}$), then read the ACF
- An AI answer is a draft: check what is stationary, what "significant" means, and whether squares or levels were tested

## After the seminar

- Lecture 1 develops each topic of today: stationarity, white noise and the random walk, Wold, ergodicity, ACF and PACF, the portmanteau tests, transformations
- C1 can grow into a team project: inflation persistence in Romania and in the region (Hungary, Poland, Czechia)
- Reading: Huang and Petukhina (2022), Ch. 1–2; Hyndman and Athanasopoulos (2021), Ch. 2–3; Brockwell and Davis (2016), Ch. 1

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